Underwater Acoustic Channel Modeling using Bellhop Ray Tracing Method

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***Abstract*—In recent years, underwater acoustic technology has rapidly developed in the mean of civil and military usage. The majority of models in underwater communication undergo the following factors those are limited bandwidth, high attenuation, time varying, multipath propagation, dispersion of signal and slow rate signal transmission. In this study we have design a GUI to examine the behavior of an acoustic signal with transmission frequency band (9 K to 90 KHz) under the above mentioned factors. We have selected two different channels environments in Arabian Sea to show channel effect over an acoustic signal, BELLHOP tool is used to design underwater channel.**

I. INTRODUCTION

Oceanic channels are considered one of the most complex medium for any transmission. Acoustic signals are best suited for underwater transmission and transmitter design. Acoustic signals are subjected to various losses which may include but not limited to absorption, scattering, multipath propagation, Doppler shifts etc. Experimental modeling of underwater channels is carried out in order to estimate transmission pattern and distance travelled by acoustic signal in sea water. To deal with aquatic channels complexity certain assumptions are required to make so that effectiveness of transmitter design can be investigated on different operational modes and frequencies tones. Keeping in view the set of assumption, pattern of acoustic signal in underwater channel, variation of the same channel effects on different frequencies bands and operating modes has made the problem significantly challenging. In the presented work, underwater acoustic channel has been modeled using BELLHOP ray tracing tool. The research work is a combination of understanding underwater acoustic propagation though theoretical and practical approaches. In theoretical part, study has been carried out related to underwater channel modeling and effects of underwater environment over the acoustic signal. In other part, MATLAB graphical user interface (GUI) has been developed for tracing different scenarios of signals using BELLHOP tool. In this work we focused to obtain two channels with high frequencies (9 to 90 KHz) acoustic propagation in deep sea for medium range (<10Km) communication and studied the behavior of sound propagation in different environment conditions, we took two propagation points in Arabian sea to model two environmental channels, channel one has a sloppy surface of 10Km, average depth of sloppy range is 180 meters with depth deviation of 275m, the second channel has flat surface of 12Km range with 52m depth . Since the band is too large to execute the behavior of channel, In order to perform ray tracing we have divide (9 to 90k Hz) band into four different frequency bands namely Extended Low frequency band (9 to 16 kHz), Low frequency band (16 to 32 kHz), Mid frequency band (31 to 53 kHz) and high frequency band (50 to 90 kHz). The simulation using MATLAB GUI has been carried out with various operational scenarios for generating different tones/ bands of frequencies. This work will provide the basic understanding of oceanic dynamics in terms of various channel losses, the design approach for underwater acoustic transmission and MATLAB simulation for effective operational requirements.

II. BELLHOP TOOL:

BELLHOP is a highly efficient ray tracing tool, which is designed to perform acoustic ray tracing for a given sound speed profile c(z) or a given sound speed field c(r, z), in ocean waveguides with flat or variable absorbing boundaries. Output options include ray coordinates, travel time, amplitude, eigenrays, acoustic pressure or transmission loss (either coherent, incoherent or semi-coherent).[2][8][9]
Input of BELLHOP is environmental file (.env) to carryout ray tracing and transmission loss, the inputs are given in table 1 that we choose to design .env, where we have shown the parameters used for modeling

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two different channels .env files to analyze the behavior of transmission loss and ray tracing.

TABLE I
PARAMETERS USED FOR CHANNEL MODELING

Parameter	Channel 1	Channel 2
Source data		
Num of sources	1	1
Source depth	25m	52m
Source frequency	Scenario dependent	Scenario dependent
Source angle	-70o to +70o	-70o to +70o
Receiver data		
Receiver ranges	[0 10]	[0 12]
Num of receiver ranges	200	200
Receiver depths	[0 400]	[0 78]
Num of receiver depths	500	500
Num of rays	100	100
Box Depth	400m	100m
Box Range	10Km	12Km
Options		
Method of interpolation	'CVW'	'CVW'
Type of media	'A*'	'A*'
*Type of information required	'C'	'C'

**To change get the plot of ray coordinates type of information can be changed into 'R', for eigenrays plot 'E' and 'C' for coherent acoustic pressure.*

III. ACOUSTIC WAVE ANALYSIS USING MATLAB

GUI has provided a set of varying parameters to depict the behavior of acoustic channel with respect to selected environmental conditions; those changeable parameters are source depth, number of sources, source angle, receiver depth, receiver range, number of receivers, and number of rays to be traced.
To investigate the acoustic signal we have designed four different modes of signals those are noise signal, pulse signal, alternate signal and sweeping signal.

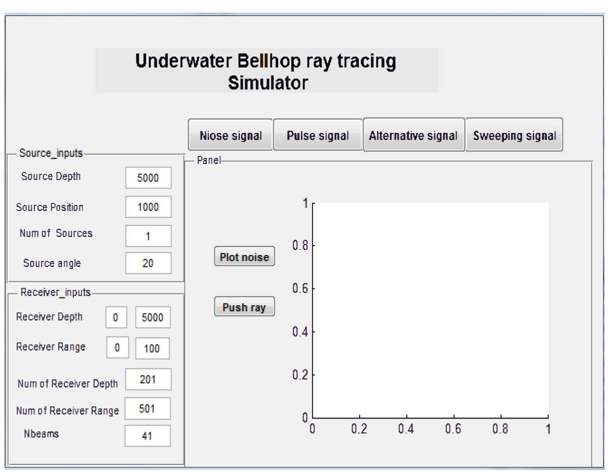

Fig. 1 GUI for underwater ray tracing using BELLHOP

A. Noise Signal:

In noise signal we generated a band limited continuous random with range of 9k - 90 kHz.

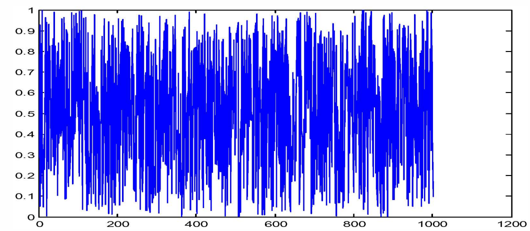

Fig 2: Band limited Noisy signal

B. Pulse Signal:

While executing the pulse signal it generates a selectable random noise with on off pulses, the width and the noise band can be vary by the user.

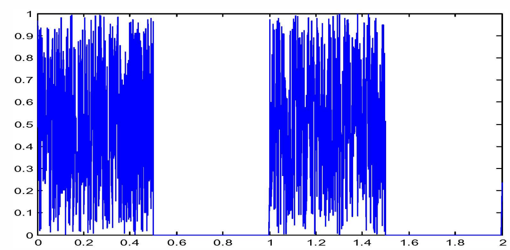

Fig 3: Band limited Noisy signal in on and off pulses

C. Alternate Signal

In this scenario convolution of multiple frequencies occurs, alternative signal is the output of four frequencies which are convoluted with a continuous random noise, these frequencies are user selectable.

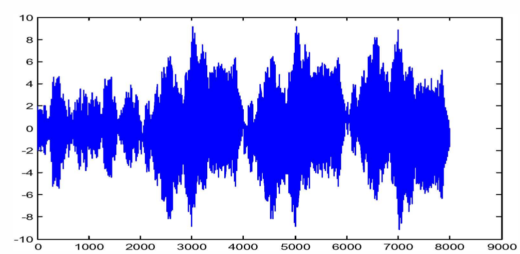

Fig 4: Alternative signal

D. Sweeping Signal:

Sweeping signal is designed by cascading multiple frequencies in one signal, where each frequency has random time slot.

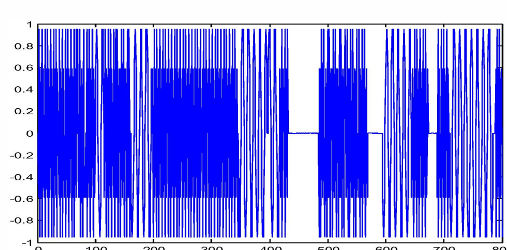

Fig 5: Sweeping frequencies signal

IV. BELLHOP IMPLEMENTATION:

The transmission losses are dependent on selected frequency, thus in order to perform simulation for a small band, BELLHOP consider center frequency for the configuration, using the arrival time and transmission loss (TL) of center frequency the behavior of complete small band can be estimated. Using the same trend we divided our large 9k to 90k Hz frequency band into four small bands namely Extended Low frequency band (9 to 16 kHz), Low frequency band (16 to 32 kHz), Mid frequency band (31 to 53 kHz) and high frequency band (50 to 90 kHz) to obtain the complete behavior of overall band using two different channel environments.

These frequencies are then provided to BELLHOP tool to calculate the transmission losses and ray tracing [11][13]. Following are the transmission losses result for channel 1 and 2; the graphs are showing the effect of channels on selected frequencies.

A. Extended Low Frequency Band:

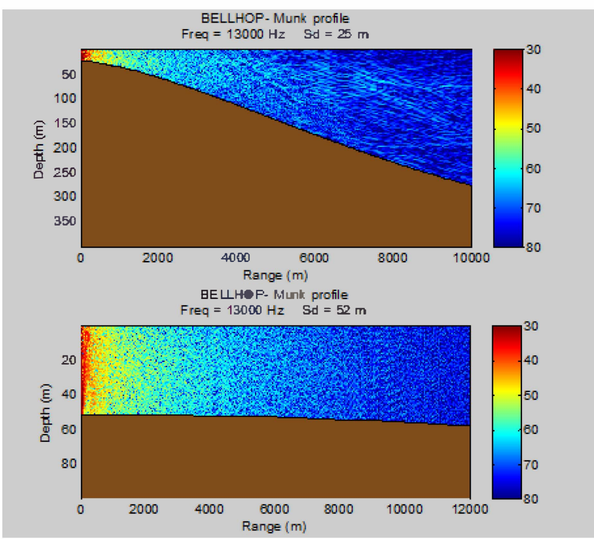

Fig 6: Channel 1 and 2 pressure filed for extended low frequency

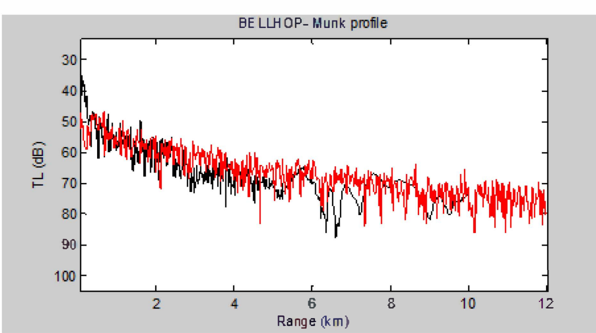

Fig 7: Channel 1 and 2 range transmission losses for extended low frequency

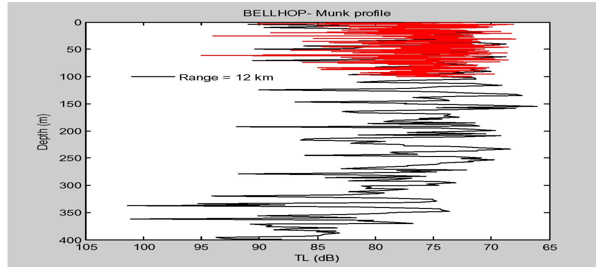

Fig 8: Channel 1 and 2 depth transmission losses for extended low frequency

To obtain the transmission losses results for extended low frequency band (9k to 16k Hz), we selected the center frequency.

Referring to fig 7, the response of range transmission losses for sloppy channel is more rapid and sharp changes can be seen which is from 30db to 88db. For the flat region channel the changes are less rapid and response of losses is more gradual, the range of losses for this channel is from 50 db to 80 db.

For depth transmission losses channel 1 has more rapid losses as compared to channel 2 the range of losses for channel 1 is from 66db to 102db, and for channel 2 the losses are from 69db to 95db.

B. Low Frequency Band:

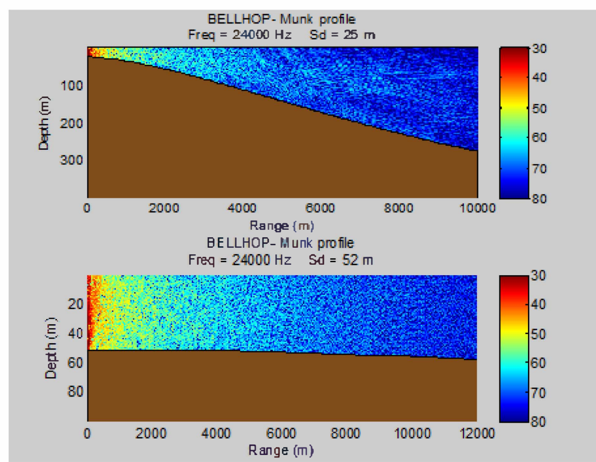

Fig 9: Channel 1 and 2 pressure filed for low frequency band

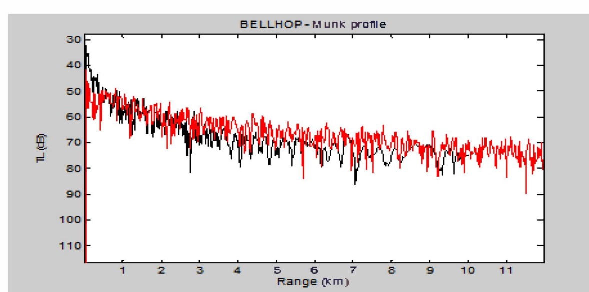

Fig 10: Channel 1 and 2 range transmission losses for low frequency

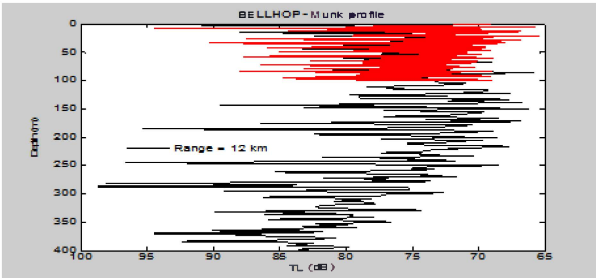

Fig 11: Channel 1 and 2 depth transmission losses for low frequency

Through fig 10 and 11 we compared the depth and range transmission losses of low frequency band (16k to 32k Hz) for channel 1 and 2, for channel 1 the transmission losses of range is from 32db to 85db there are less numbers of sharp dips as we observed in case one, for channel 2 sharp dips are more and the range of losses is 42db to 90db.

Again For depth transmission losses channel 1 has more rapid losses as compared to channel 2 but the number of pulse has increased in channel 2 by the last scenario the range of losses for channel 2 is from 65db to 94.8db, and for channel 1 the losses are from 65.5db to 99db.

C. Mid Frequency Band:

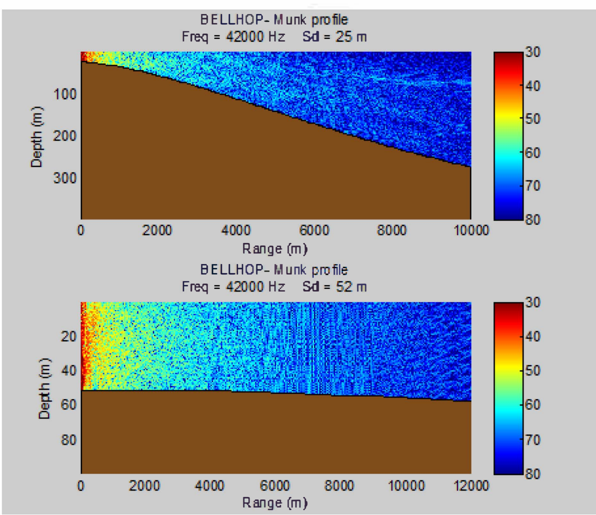

Fig 12: Channel 1 and 2 pressure filed for mid frequency band

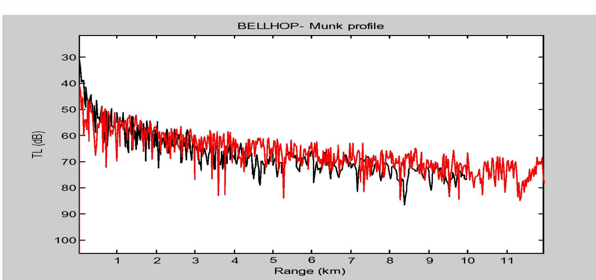

Fig 13: Channel 1 and 2 range transmission losses for low frequency band

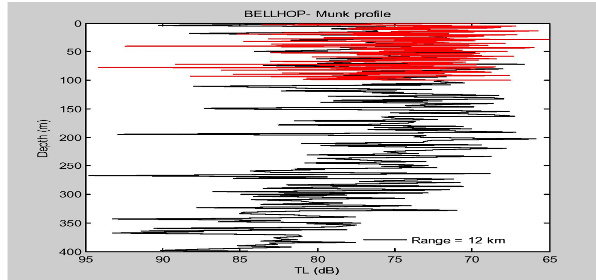

Fig 14: Channel 1 and 2 depth transmission losses for low frequency band

In this simulation we compared the depth and range transmission losses of medium frequency band (31 to 52k Hz) for channel 1 and 2.

Fig 13 and 14 are showing the depth and range transmission losses for channel 1 and 2, the transmission losses of range for channel 1 is from 32db to 80db the losses are increasing there are less numbers of sharp dips as compared to channel 2, for channel 2 range of losses is 42db to 83db. For depth transmission losses channel 1 has losses from 65.9db to 94.9db and for channel 2 is from 65db to 94.db.

D. High frequency Band :

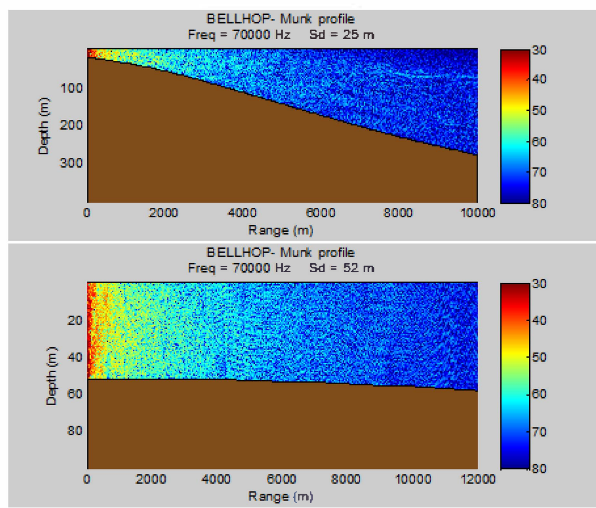

Fig 15: Channel 1 and 2 acoustic pressure filed for high frequency

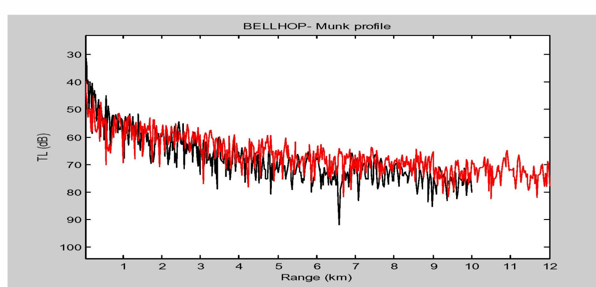

Fig 16: Channel 1 and 2 range transmission losses for high frequency

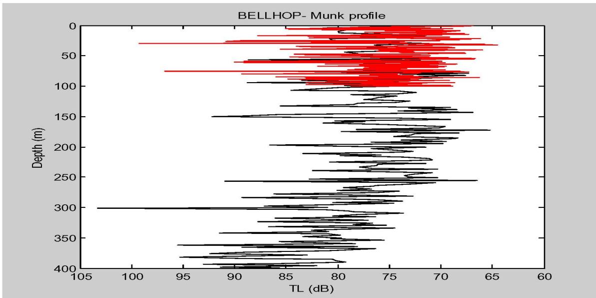

Fig 17: Channel 1 and 2 depth transmission losses for high frequency band

In the last simulation we compared the results of high frequency band (50 to 90k Hz) where Fig 16 and 17 are representing the depth and range transmission losses for channel 1 and 2.

As per the fig16 channel 1 has transmission losses from 34db to 90db and for channel 2 40db to 70 db. Depth losses for channel 1 is from 65 to 95 db but there is a sharp pulse at 300 depth of 104 db, for channel 2 range of losses is from 65db to 99db

V. RESULTS

Through comparison of all four small bands on the basis of selected channels we have observed, since channel 1 has less depth at source point which cause low transmission losses when signal is near to source, as the depth increases transmission losses also tends to increase, but for channel 2 the depth of bottom surface is more at source point compared to channel 1 which cause more transmission losses at the start, since the depth is constantly same so the overall losses are lesser for channel 2 compared to channel 1.

TABLE II

	Ex. Low frequency band	Low frequency band	Mid frequency band	High frequency band
Channel 1 (depth vs. TL)	66 to 102db	65 to 99db	65.9 to 94.9db	65 to 95db
Channel 2 (depth vs. TL)	69 to 95db	65 to 94.8db	65 to 94db	65 to 99db
Channel 1 (range vs. TL)	30 to 88db	32 to 85db	32 to 80db	34 to 90db
Channel 2 (range vs. TL)	50 to 80db	42 to 90db	42 to 83db	40 to 70db

Table 2 summarize the analysis we have done over multiple frequency bands, different transmission losses are found at various points, by making a comparison between two channels for short range communication and we have determine that, channel 1 has shown better performance for low frequency bands as compared to high frequencies.

For channel 2 since the channel has flat surface which cause gradual increase in losses but the depth of the channel is very low as compared to channel 1 which results more numbers of reflections and causes sharp loss at some locations. For mid and high frequency channel 2 has shown better performance compared to channel 1.

VI. CONCLUSION

The objective was to design a dynamic platform where using different oceanic condition we would be able to examine an acoustic transmission. The analysis work revealed that more efficiency can be achieved by selecting appropriate components for sound propagation in different scenarios, the work can be utilized for variety of applications including masking of platform signatures by deploying false acoustic tones, this study can be the basis for various underwater acoustic analysis work and defense applications etc.

REFERENCES

[1] J. Peterson and M. Porter, "Ray/Beam Tracing for Modeling the Effects of Ocean and Platform Dynamics", *IEEE Journal of Oceanic Engineering*, vol. 38, no. 4, pp. 655-665, 2013.

[2] Porter M.B. and Bucker H.P. Gaussian beam tracing for computing ocean acoustic fields. J. Acoust. Soc. America, 82(4):1349–1359, 1987.

[3] Jensen F., Kuperman W., Porter M., and Schmidt H. Computational Ocean Acoustics. AIP Series in Modern Acoustics and Signal Processing, New York, 1994.

[4] A. MacGillivray, "Modeling underwater sound propagation from an airgun array using the parabolic equation method", *The Journal of the Acoustical Society of America*, vol. 122, no. 5, p. 2942, 2007.

[5] N. Chotiros and M. Stern, "High-frequency acoustic wave propagation modeling in shallow water",*The Journal of the Acoustical Society of America*, vol. 100, no. 4, p. 2701, 1996.

[6] R. Frosch, "Underwater Sound: Deep-Ocean Propagation: Variations of temperature and pressure have great influence on the propagation of sound in the ocean.", *Science*, vol. 146, no. 3646, pp. 889-894, 1964.

[7] Milica Stojanovic, "Underwater Acoustic Communication Channels: Propagation Models and Statistical Characterization.", *IEEE Communications Magazine*, January 2009

[8] M. B. Porter, Ocean Acoustics Library, Bellhop, http://oalib.hlsresearch.com/.

[9] Porter M.B. The BELLHOP Manual and User's Guide

[10] P. H. Milne, Underwater acoustic positioning systems. E. & F.N. Spon Ltd, London, 1983.

[11] "Bellhop gaussian beam/finite element beam code," Available in the Acoustics Toolbox, http://oalib.hlsresearch.com/Rays, 2015.

[12] R. Thiele, "NATO model study," SACLANT Undersea Research Rep., SR-149, 1989.

[13] M. B. Porter and E. L. Reiss, "A numerical method for Ocean acoustic normal modes," J. Acoust. Soc. of Am., vol. 76, no. 1, pp. 244-252, 1984.

[14] D. Lee, G. Botseas, and J. S. Papadakis, "Finite difference Solutions to the parabolic equation," J. Acousr. Soc. Am., vol. 70, pp. 795-800, 1981.

[15] F. Tappert, 'The parabolic equation method," in Wave Propagation und Undernuter Acoustics. J. B. Keller and J. S. Papadakis, Eds. New York: Springer-Verlag. 1977.